



Technical Service Bulletin

Technical Service Bulletin: TSB101114

Released Date: 06-Jan-2011

Dark Return Fuel on Engines with HPI Fuel System

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

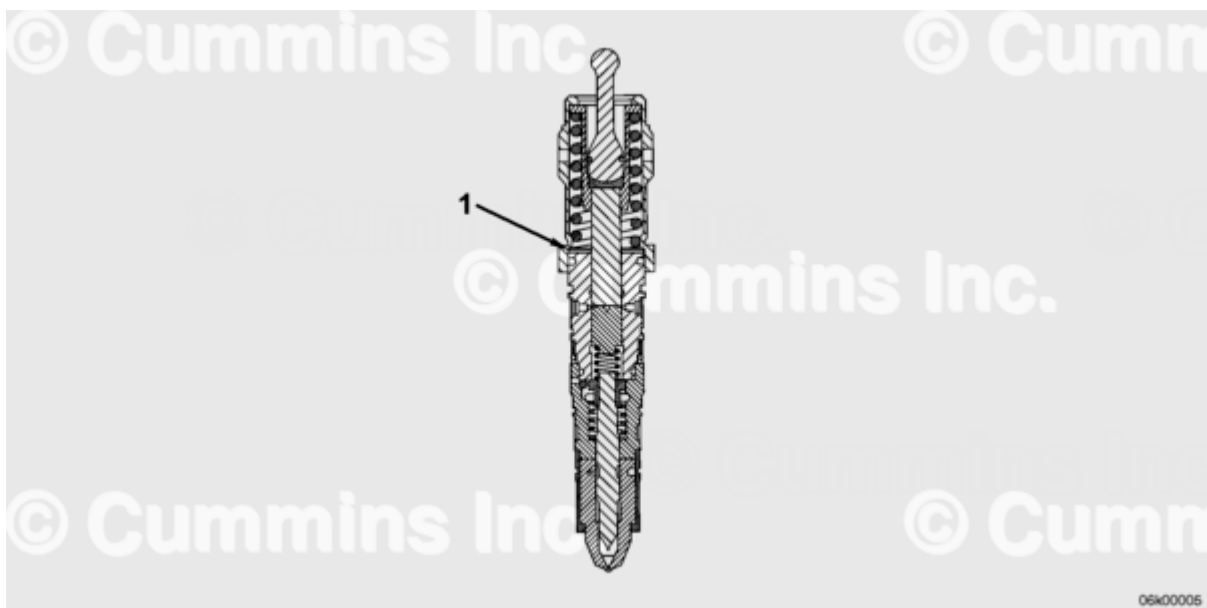
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This document addresses the issue of dark return fuel color on engines equipped with the HPI fuel system (QSK engines with mechanically actuated injectors).

Fuel returning to the tank from the fuel drain line may appear dark in color. Dark color of the return fuel does **not** imply engine malfunction, and does **not** necessarily affect engine performance.

There are two main causes for dark return fuel on HPI fuel systems:

1. Lubricating oil entering the fuel system through the fuel injector.
2. Formation of asphaltenes, due to high temperatures and pressures in the fuel injector.



1. Lubricating oil from the overhead can enter the top of the injector and get past the top injector seal (1). This allows the oil to enter the timing fuel chamber and then be returned back to the tank with the rest of the return fuel. The amount of oil allowed into the fuel system by the injector is

less than 0.1 percent of total fuel consumption. This amount of oil in the fuel does **not** affect engine or fuel filter performance, durability, or reliability.

When an engine displays dark return fuel, check the color of the engine exhaust or the amount of oil consumed. These are better indications of condition of the engine. Do **not** use the color of the return fuel as an indicator of engine condition. The color can change, depending on the amount of fuel left in the tank and the types of oil and fuel used.

2. The high fuel pressures and temperatures in the HPI fuel injector can cause asphaltene formation due to fuel overheating. The high temperature fuel that is **not** injected into the cylinder is returned to the fuel tank and appears dark in color. Asphaltene formation due to fuel overheating can be aggravated by a lack of fuel coolers, improperly sized fuel tanks, inadequate mixing of return fuel between tanks, low fuel levels, or fuel tank installation that prevents dissipation of heat.

If an engine or fuel system cooling issue is suspected, these cooling systems **must** be inspected and modified as necessary to comply with Cummins Inc. requirements.

Use Fuels for Cummins® Engines, Bulletin 3379001, for more information on fuel discoloration or asphaltene formation.

Use AEB 24.07 or contact a Cummins® Authorized Repair Location for information on fuel cooling requirements and installation guidelines.

Document History

Date	Details
2010-12-10	Module Created

Last Modified: 06-Jan-2011
